

Name: _____

Section: _____

Date: _____

SECTION A – Fill in the Blanks (1 mark each)

1. The term 'virus' was coined by _____.
2. _____ (1892) discovered the causative agent of tobacco mosaic disease.
3. _____ (1898) named the infectious agent Contagium vivum fluidum.
4. _____ (1935) crystallised viruses and showed they were largely composed of protein.
5. The protein coat surrounding a virus's nucleic acid is called the _____.
6. The subunits of the viral capsid are called _____.
7. A completely assembled virus particle found outside a host cell is called a _____.
8. Viroids were discovered by _____ in 1971.

SECTION B – One-Word / Very Short Answer (1 mark each)

1. What type of genetic material does Tobacco Mosaic Virus (TMV) contain? **Ans:** _____
2. What type of genetic material do bacteriophages typically contain? **Ans:** _____
3. Name the disease caused by viroids in potato plants. **Ans:** _____
4. Name the vector that transmits potato spindle tuber viroid. **Ans:** _____
5. Name the prion disease found in cattle. **Ans:** _____
6. Name the human disease analogous to mad cow disease. **Ans:** _____

SECTION C – Match the Following (4 marks)

Column A	Column B
1. TMV	a. Infectious protein, no nucleic acid
2. Bacteriophage	b. Double-stranded DNA
3. Viroid	c. Free RNA, no protein coat
4. Prion	d. Single-stranded RNA

Answer: 1-____, 2-____, 3-____, 4-____

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SECTION D – True or False (1 mark each)

1. Viruses contain both DNA and RNA together as genetic material. [True / False]
2. Viroids lack a protein coat and exist as free RNA. [True / False]
3. Prions are infectious protein particles causing neurological disease. [True / False]
4. A virion is a fully assembled virus particle found outside a host. [True / False]

SECTION E – Short Answer (3 marks each)

1. Differentiate between a virus and a viroid based on structure.

2. Explain why viruses are not considered truly living organisms outside a host cell.

3. Name the scientists associated with the discovery/description of TMV, viroids, and prions respectively.

SECTION F – HOTS / Long Answer (5 marks each)

1. Explain, with reasons, why prions are considered unusual infectious agents compared to viruses and viroids.

2. Trace the historical sequence of discoveries (Ivanowsky, Beijerinck, Stanley) that led to our understanding of viruses.

Teacher's Signature: _____ Date: _____

Teacher's Remarks:
